Calculation questions – 4.2 Electricity

AQA GCSE Physics (8463) Higher · Paper 1 · 17 questions · 58 marks · answers at the back

Equations for this topic

Equation	Symbols and units	Exam
Q = It	Q (C), I (A), t (s)	Learn it
V = IR	V (V), I (A), R (Ω)	Learn it
P = VI	P (W), V (V), I (A)	Learn it
P = I²R	P (W), I (A), R (Ω)	Learn it
E = Pt	E (J), P (W), t (s)	Learn it
E = QV	E (J), Q (C), V (V)	Learn it
Series: R total = R1 + R2; same current; p.d. shared		Rule
Parallel: same p.d. across each branch; branch currents add		Rule

Write the equation, substitute, then rearrange. Convert to base units first. Give a unit and 2–3 significant figures. Take g = 9.8 N/kg.

Grade 4–5

1 A current of 0.50 A flows through a 12 Ω resistor. Calculate the potential difference across it. [2 marks]

Answer: _____ Unit: _____

2 A current of 3.0 A flows for 20 s. Calculate the charge that flows. [2 marks]

Answer: _____ Unit: _____

3 A lamp on 230 V mains draws a current of 0.26 A. Calculate its power. [2 marks]

Answer: _____ Unit: _____

4 A 4.0 Ω and a 6.0 Ω resistor are connected in series. Calculate the total resistance. [1 mark]

Answer: _____ Unit: _____

5 15 C of charge passes through a p.d. of 9.0 V. Calculate the energy transferred. [2 marks]

Answer: _____ Unit: _____

Grade 6–7

6 A current of 250 mA flows when the p.d. is 6.0 V. Calculate the resistance. [3 marks]

Answer: _____ Unit: _____

7 A 2.0 kW heater runs from 230 V mains. Calculate the current. [3 marks]

Answer: _____ Unit: _____

8 A cable of resistance 0.40 Ω carries a current of 15 A. Calculate the power wasted heating the cable. [2 marks]

Answer: _____ Unit: _____

9 A phone charges at 1.2 A for 2.0 hours from a 5.0 V charger. (a) Calculate the charge that flows. (b) Calculate the energy transferred. [5 marks]

Answer: _____ Unit: _____

10 A 3.0 kW kettle is used for 4.0 minutes. (a) Calculate the energy transferred in kJ. (b) Electricity costs 30p per kWh. Calculate the cost. [5 marks]

Answer: _____ Unit: _____

11 A 12 V battery is connected to a 2.0 Ω and a 4.0 Ω resistor in series. Calculate (a) the current and (b) the p.d. across the 4.0 Ω resistor. [4 marks]

Answer: _____ Unit: _____

Grade 8–9

12 Two 6.0 Ω resistors are connected in parallel across a 3.0 V cell. (a) Calculate the total current from the cell. (b) Calculate the total resistance and explain why it is less than either resistor. [5 marks]

Answer: _____ Unit: _____

13 A 12 V battery is connected across a 6.0 Ω resistor and a 12 Ω resistor in parallel. Calculate the current from the battery and the total resistance of the circuit. [4 marks]

Answer: _____ Unit: _____

14 A thermistor is in series with a 1.5 kΩ fixed resistor across a 9.0 V supply. The p.d. across the fixed resistor is 3.0 V. Calculate the resistance of the thermistor. [4 marks]

Answer: _____ Unit: _____

15 A power station sends 50 MW through cables of total resistance 5.0 Ω. Calculate the power lost in the cables if it is transmitted at (a) 25 kV and (b) 400 kV. (c) Use your answers to explain why the National Grid uses step-up transformers. [6 marks]

Answer: _____ Unit: _____

16 A current of 0.32 A flows in a wire. The charge on one electron is 1.6×10^{-19} C. Calculate how many electrons pass a point each second. [3 marks]

Answer: _____ Unit: _____

17 A kettle element is rated 2.3 kW at 230 V. (a) Calculate its resistance. (b) The supply drops to 220 V. Calculate the new power, assuming the resistance stays the same. [5 marks]

Answer: _____ Unit: _____

Answers – 4.2 Electricity

Mark yourself honestly: a mark for the right equation, a mark for substitution, a mark for the answer with its unit.

Q	Working	Answer
1	V = 0.50 × 12	**6.0 V**
2	Q = 3.0 × 20	**60 C**
3	P = 230 × 0.26	**60 W**
4	4.0 + 6.0	**10 Ω**
5	E = 15 × 9.0	**135 J**
6	R = 6.0 ÷ 0.250	**24 Ω**
7	I = 2000 ÷ 230	**8.7 A**
8	P = 15^2 × 0.40	**90 W**
9	(a) Q = 1.2 × 7200 (b) E = 8640 × 5.0	**(a) 8640 C (b) 43 200 J**
10	(a) 3000 × 240 = 720 000 J (b) 3.0 kW × (4 ÷ 60) h = 0.20 kWh; × 30p	**(a) 720 kJ (b) 6p**
11	R = 6.0 Ω; I = 12 ÷ 6.0; V = 2.0 × 4.0	**(a) 2.0 A (b) 8.0 V**
12	each branch 3.0 ÷ 6.0 = 0.50 A; total 1.0 A; R = 3.0 ÷ 1.0. A second branch gives the current an extra path, so more current flows for the same p.d.	**(a) 1.0 A (b) 3.0 Ω**
13	12 ÷ 6.0 = 2.0 A; 12 ÷ 12 = 1.0 A; total 3.0 A; R = 12 ÷ 3.0	**3.0 A, 4.0 Ω**
14	I = 3.0 ÷ 1500 = 0.0020 A; thermistor p.d. = 9.0 − 3.0 = 6.0 V; R = 6.0 ÷ 0.0020	**3000 Ω (3.0 kΩ)**
15	(a) I = 50×10^6 ÷ 25×10^3 = 2000 A; P = 2000^2 × 5.0 (b) I = 125 A; P = 125^2 × 5.0 (c) higher p.d. → lower current; loss ∝ I^2	**(a) 2.0 × 10^7 W (40%) (b) 7.8 × 10^4 W**
16	0.32 ÷ 1.6 × 10^{-19}	**2.0 × 10^{18}**
17	(a) I = 2300 ÷ 230 = 10 A; R = 230 ÷ 10 (b) I = 220 ÷ 23 = 9.57 A; P = 220 × 9.57	**(a) 23 Ω (b) 2.1 kW**

Common mistakes. Not converting mA and kΩ (Q6, Q14); using hours instead of seconds in Q = It (Q9); adding parallel resistances like series ones (Q12).